Aligning Text/Speech Representations from Multimodal Models with MEG Brain Activity During Listening

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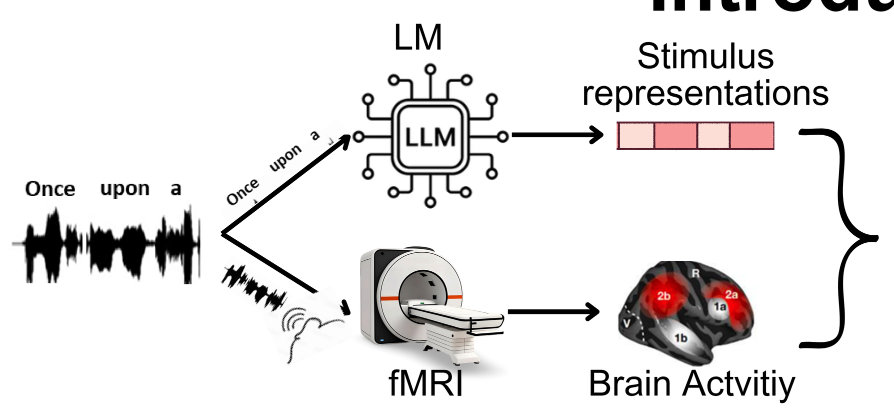
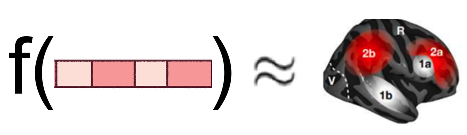
EMNLP 2025 Suzhou, China 中国苏州

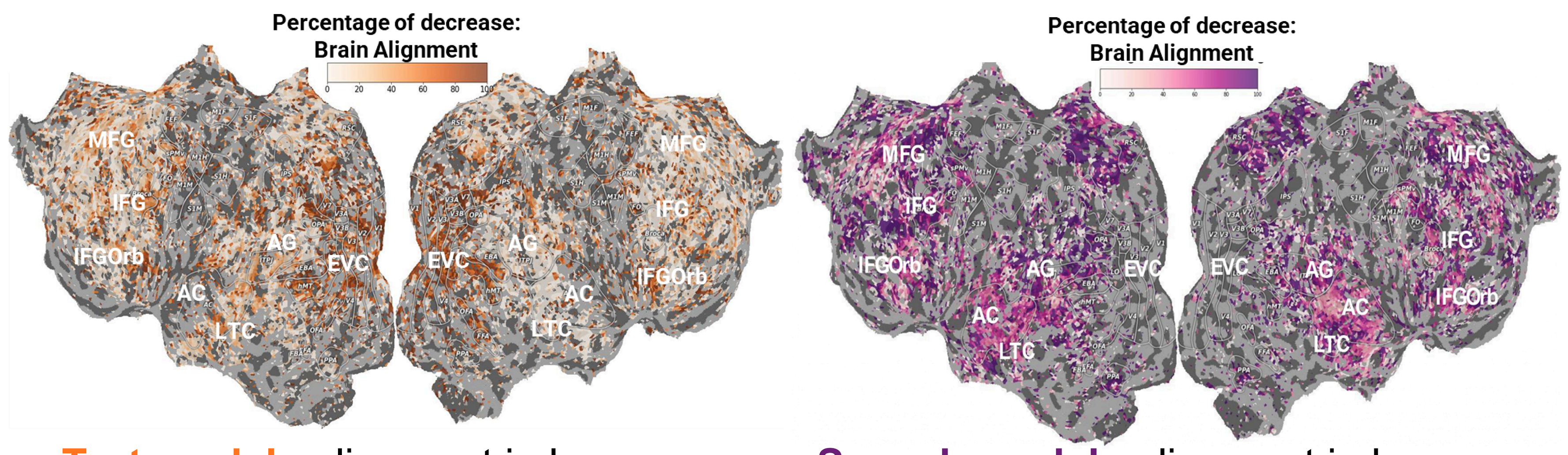
Introduction



LMs predict brain activity when participants listen to or read a complex language stimulus [1,2]



Brain alignment of a LM ⇒ how similar its representations are to a human brain



Text models alignment in language regions is due to brain-relevant semantics

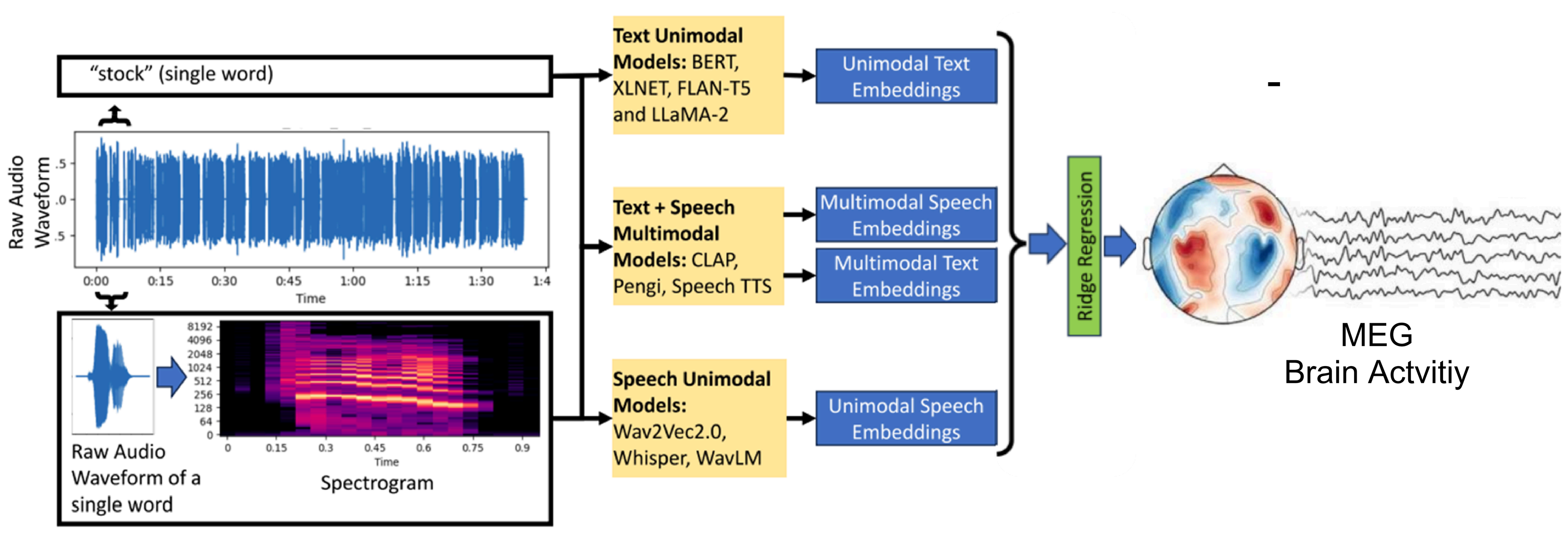
Speech models alignment in language regions is due to low-level features

Multimodal (Text+Speech) models enables learning audio concepts from natural language supervision

"Can <u>speech embeddings</u> from <u>multimodal models</u> capture brain-relevant semantics through cross-modal interactions?"

Methods

Multi-modal vs. Unimodal models: Brain alignment

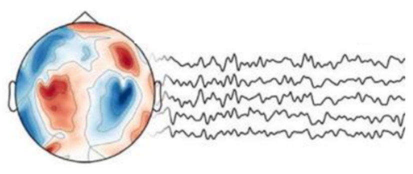


Which modality of representations in multi-modal models leads to high brain alignment?

Residual Method [3]

Step 1 $g(X_1) \approx X_2$ ⟶ g: linear regression estimating X_2 from X_1; X_1: Unimodal Speech Embeddings; X_2: Multimodal Text Embeddings

Step 2 $|X_2 - g(X_1)|$ = Residuals

Step 3 $h(|X_2 - g(X_1)|) \approx$ 

Brain Dataset [4]

Dataset	Modality	#Subj	Language	#stories
MEG-MASC	Listening	27	English	4 stories

Pretrained Transformer models

Text+Speech Multimodal	CLAP, SpeechT5, Pengi
Unimodal Text	BERT, XLNet, FLAN-T5, LLaMA-2
Unimodal Speech	Wav2Vec2.0, WavLM, Whisper

References

[1] Deniz et al. 2019. The Representation of Semantic Information Across Human Cerebral Cortex During Listening Versus Reading Is Invariant to Stimulus Modality. The Journal of neuroscience

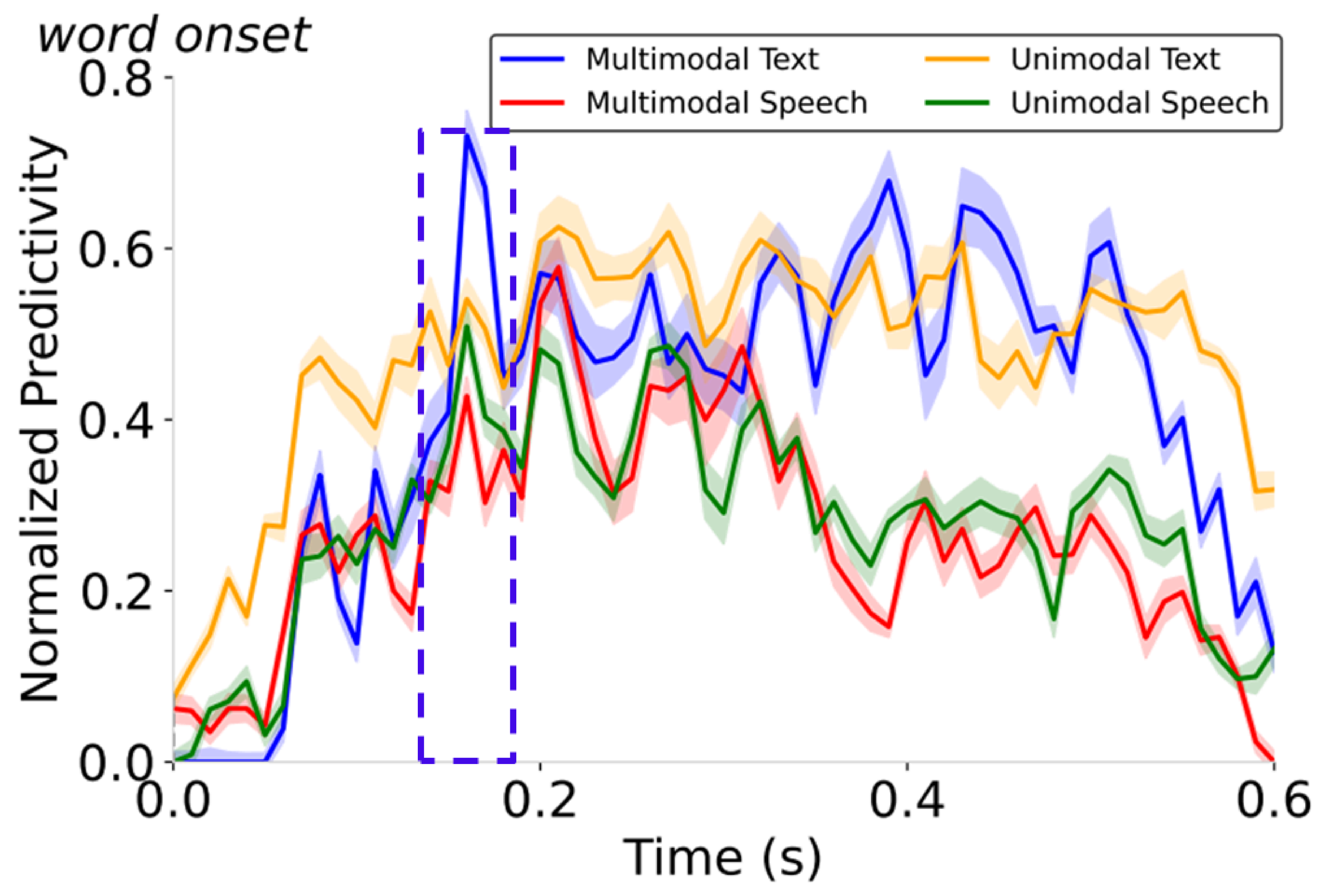
[2] Oota et al. 2024. Speech language models lack brain-relevant semantics

[3] Toneva et al. 2022. Combining computational controls with natural text reveals aspects of meaning composition. Nature computational science, 2(11), 745–757.

[4] Gwilliams et. al 2022. Introducing meg-masc a high-quality magneto-encephalography dataset for evaluating natural speech processing. Scientific data

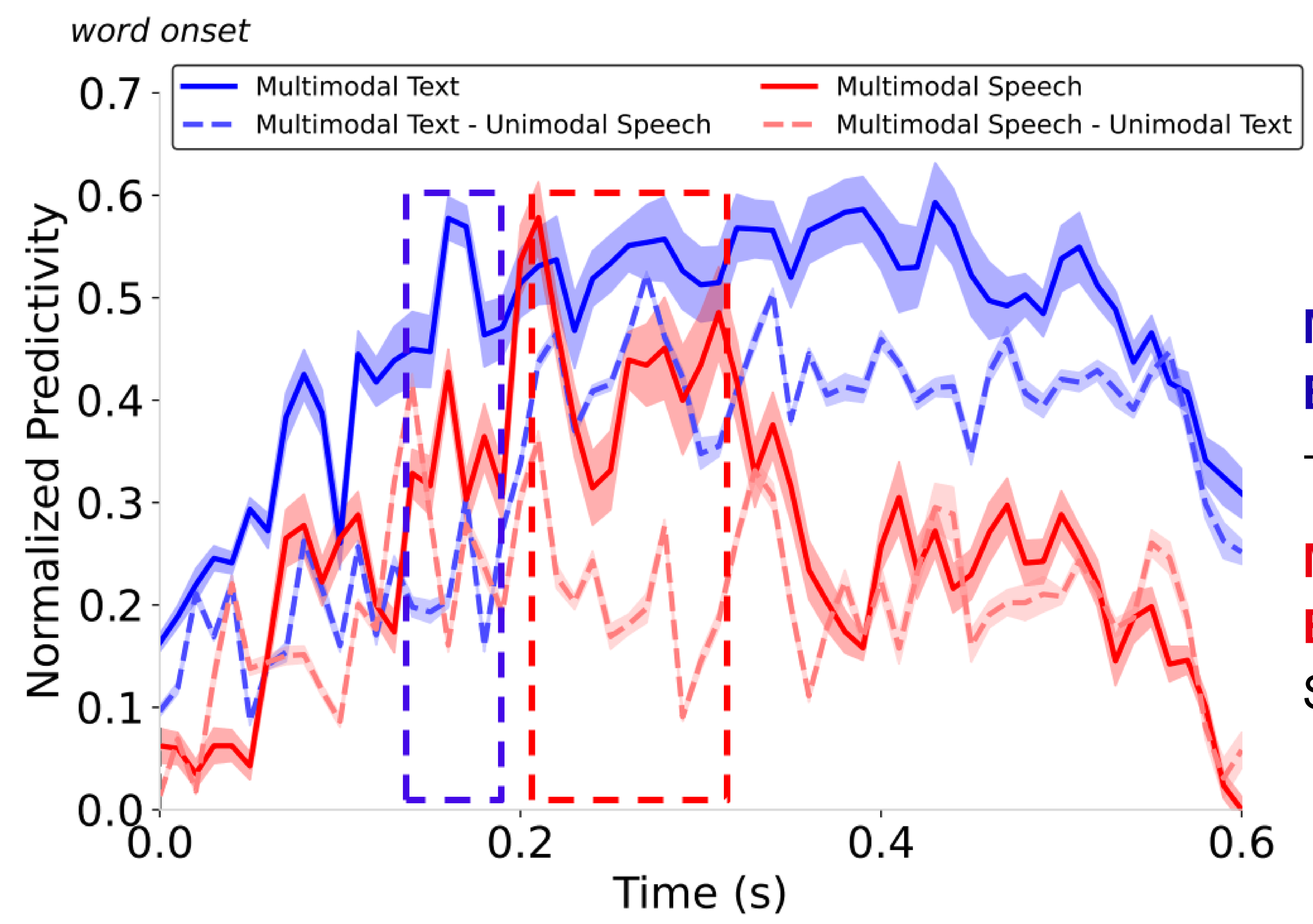
Results

Multimodal Text Embeddings benefit from speech modality



Multimodal Text Embeddings:
- peak around 200 ms
- benefit from speech modality.

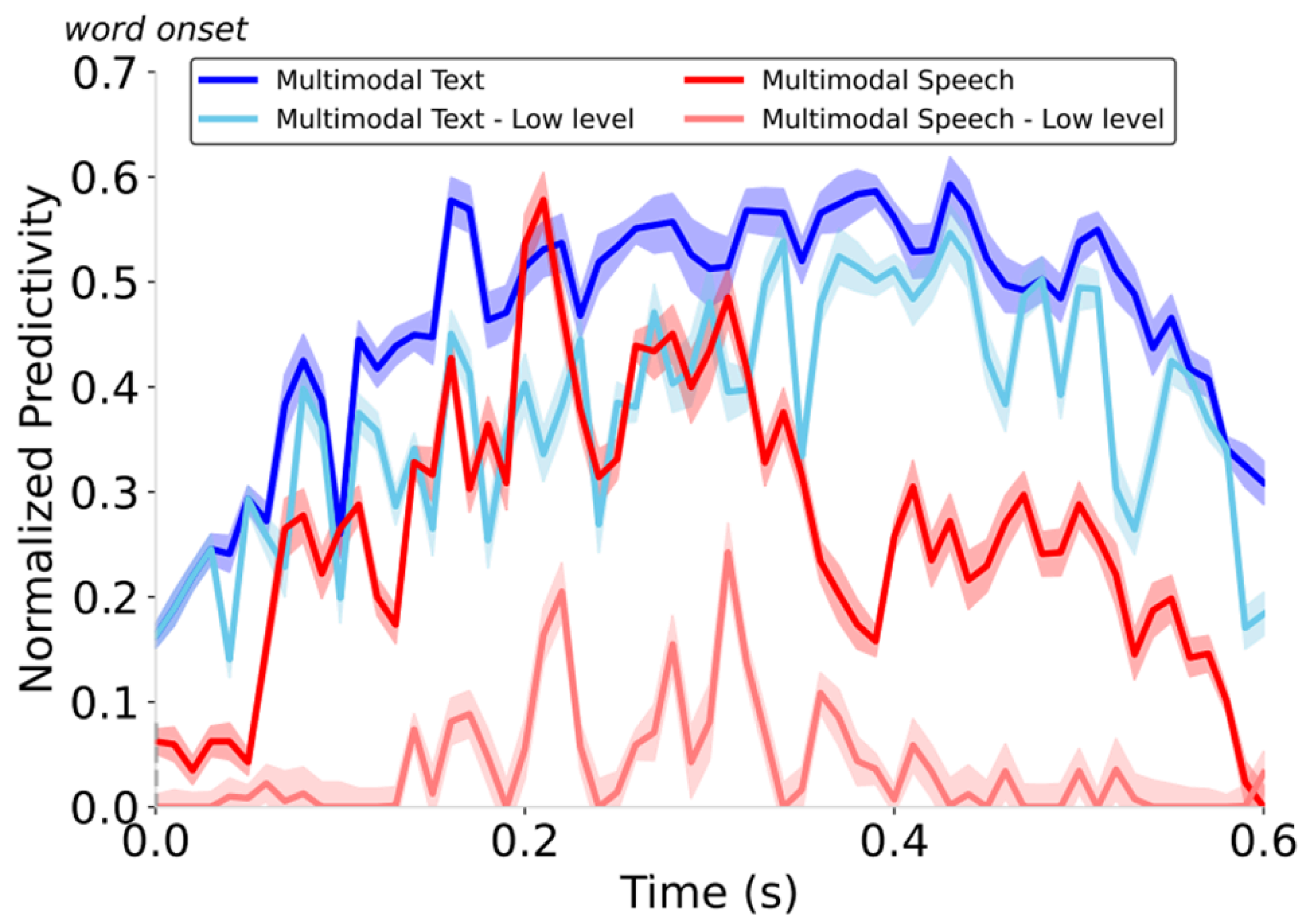
Multimodal models show asymmetric cross-modal integration



Multimodal Text Embeddings: Speech → Text: acoustics

Multimodal Speech Embeddings: Text → Speech: lexical

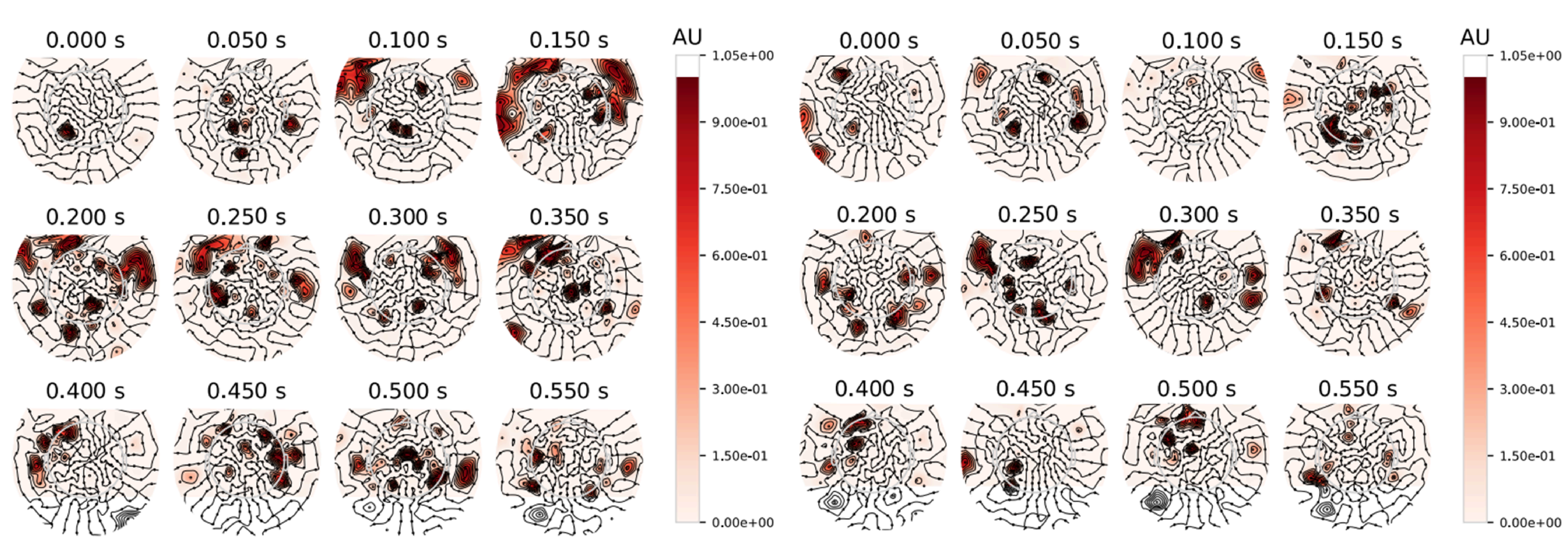
Multimodal speech embeddings lack brain-relevant semantics



Multimodal Text Embeddings: high brain alignment due to brain-relevant semantics

Multimodal Speech Embeddings: brain alignment mostly due to low-level speech features

Qualitative analysis: Topographic maps



(a) Multimodal Text - Lowlevel (b) Multimodal Speech - Lowlevel

Conclusion

- **Multimodal Speech models** are useful for modeling early auditory and lexical word processing. Need to investigate speech models to learn more about word processing
- **Multimodal Text models** are useful for modeling both early auditory and high-level semantic information processing
- More work to do for a complete end-to-end multimodal model capable of bi-directional cross-knowledge transfer between and